

Unified Cauer Ladder + Surface Impedance Network via Turning-Point Basis Switch

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Abstract—A truncated Cauer Ladder Network (CLN) is seamed to a Surface Impedance Boundary Condition (SIBC) element ($\alpha = 1/2$ Constant Phase Element) at a turning point ω_N given by Matsuo’s corner-frequency formula, with the SIBC scaling fixed by continuity (no fitting). Formally this is the s_∞ expansion point of the multiple-expansion-points CLN framework. Analytical verification on a Cu slab reproduces the Stoll/Wall closed form across nine decades with ten Cauer rungs and one SIBC element; on a 3D Cu cuboid, an mpmath single-element VIM with HDiv divergence-free hex basis and an NGSolve HCurl Tanimoto-AT extraction yield matching unified $Y(j\omega)$ from two bulk Cauer rungs plus the SIBC element.

Index Terms—Cauer Ladder Network, Surface Impedance, Turning Point, Expansion Points, Eddy Current.

I. UNIFIED CLN + SIBC

A truncated Cauer Ladder Network (CLN) [1] represents the eddy-current admittance $Y(s) = 1/Z(s)$ as a continued fraction with rungs (R_{2k}, L_{2k+1}) encoding the Foster eddy-mode spectrum. The reliable rung count is bounded both by Hankel-Pad  ill-conditioning ($\sim 300\times$ noise amplification per stage [7]) and, more fundamentally, by the basis’s spatial resolution – once skin depth $\delta(\omega) = \sqrt{2/(\omega\mu\sigma)}$ drops below the basis length scale, further rungs are non-physical. In that regime, however, the Surface Impedance Boundary Condition $Z_s(j\omega) = \sqrt{j\omega\mu/\sigma}$ – a Constant Phase Element of exponent $\alpha = 1/2$ – becomes exact.

We seam the two representations at a turning point ω_N given by Matsuo’s corner-frequency formula [3] evaluated at the last reliable rung,

$$\omega_N = \sqrt{2^{\alpha/2} - 1} R_{2N-2}/L_{2N-1} \approx 0.4350 R_{2N-2}/L_{2N-1}, \quad (1)$$

yielding $Y_{unified}(j\omega) = 1/Z_{Cauer}^{(N)}(j\omega)$ for $\omega < \omega_N$ and $K_{SIBC} \sqrt{\sigma/(j\omega\mu)}$ above, with $K_{SIBC} = Y_{Cauer}^{(N)}(j\omega_N)/\sqrt{\sigma/(j\omega_N\mu)}$ fixed by continuity (no fitting). This extension is the s_∞ expansion point of the multiple-expansion-points CLN framework [2]: $K_\infty/s_\infty \rightarrow \sigma$ degenerates the energy operator to the conductivity inner product, the natural basis for surface-localised modes.

II. VERIFICATION

Slab. For a Cu slab of half-thickness $c = 1$ mm, Stoll’s $Y(s) = \sigma c \tanh(u)/u$ with $u = c\sqrt{s\mu\sigma}$ [4] and Wall’s odd-integer continued fraction [5] provide the analytical ladder. Feeding the Mittag-Leffler Foster spectrum into our mpmath QD-Wynn extractor at 60-digit precision recovers ten rungs with $k_L = k_R = 0$ [3] and $\alpha = 0.500$ to all reported digits; ω_N at $N = 10$ gives $f_N = 1.12$ MHz, $K_{SIBC} = 0.937$, and $Y_{unified}$ matches Y_{full} to under 10% at the seam and 1% in the deep Cauer/SIBC regimes across DC to 10^8 Hz.

3D Cu cuboid (17.72×17.72×2 mm). A pure-mpmath single-element Volume Integral Method (VIM) with HDiv divergence-free hex basis [6] at $p=3$ (81 dof) produces three Cauer rungs $\tau_{pair} = 194.66, 69.96, 19.40 \mu s$, of which the third lies in the basis noise floor; the *mesh-natural* truncation retains two rungs and seams the SIBC element at $f_N = 9.9 \times 10^2$ Hz. An NGSolve HCurl Tanimoto-AT extraction at $p=3$, $h=0.5$ mm produces an extra leading stage corresponding to the $t=0^+$ skin-layer

impulse response; identifying it as the SIBC seed and using stages $\{1, 2\}$ as bulk rungs gives $\tau_{pair} = 218.99, 84.16 \mu s$, within 12–20% of VIM. The unified $Y(j\omega)$ from the two independent extractions overlap to graphical precision (Fig. 1). A direct NGSolve frequency-domain AC sweep ($A_{red} \cdot t = 0$ Dirichlet, 20 frequencies 1 Hz– 10^7 Hz) provides the ground-truth $Y_{AC}(j\omega) = |A_{ext}|^2 + \langle A_{red}, A_{ext} \rangle$, which coincides with the VIM/FEM unified prediction throughout the Cauer regime (below ~ 1 kHz) and decays slightly faster than the SIBC seam above it, consistent with the conductor’s finite-size Faraday-shielding floor.

REFERENCES

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A1 cuboid 17.72×17.72×2 mm Cu — Cauer + SIBC unified $Y(2s)$

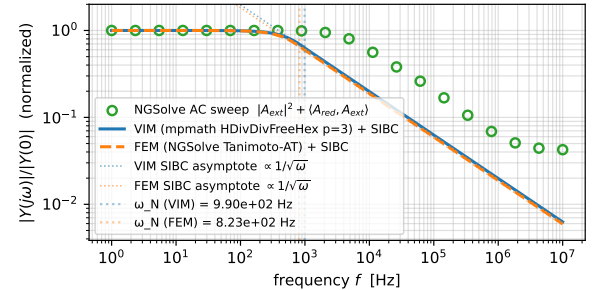


Fig. 1. Cauer + SIBC unified $|Y(j\omega)|/|Y(0)|$ for the 17.72 × 17.72 × 2 mm Cu cuboid in B_z . VIM (solid) and NGSolve Tanimoto-AT (dashed) overlap across nine decades. Green circles: independent NGSolve AC sweep ground truth – matches the unified prediction in the Cauer regime and lies slightly below the SIBC seam at high ω owing to finite-conductor Faraday shielding.